

In []: *#EXPERIMENT NO. 9*

```
import pandas as pd
import matplotlib.pyplot as plt

# Load Dataset
df = pd.read_csv("superstore.csv", encoding="latin1")

print("Dataset Loaded Successfully")
print("Dataset Shape:", df.shape)

# Sales by Category
category_sales = df.groupby("Category")["Sales"].sum()

plt.figure(figsize=(6,4))
plt.bar(category_sales.index, category_sales.values)
plt.title("Sales by Category")
plt.xlabel("Category")
plt.ylabel("Sales")
plt.show()

# Sales by Region
region_sales = df.groupby("Region")["Sales"].sum()

plt.figure(figsize=(7,4))
plt.bar(region_sales.index, region_sales.values)
plt.title("Sales by Region")
plt.xlabel("Region")
plt.ylabel("Sales")
plt.xticks(rotation=45)
plt.show()

# Sales by Segment
segment_sales = df.groupby("Segment")["Sales"].sum()

plt.figure(figsize=(6,4))
plt.bar(segment_sales.index, segment_sales.values)
plt.title("Sales by Segment")
plt.xlabel("Segment")
plt.ylabel("Sales")
plt.show()

# Top 10 Countries
country_sales = df.groupby("Country")["Sales"].sum().sort_values(ascending=False).h

plt.figure(figsize=(8,5))
plt.barh(country_sales.index, country_sales.values)
plt.title("Top 10 Countries by Sales")
plt.xlabel("Sales")
plt.ylabel("Country")
plt.gca().invert_yaxis()
plt.show()

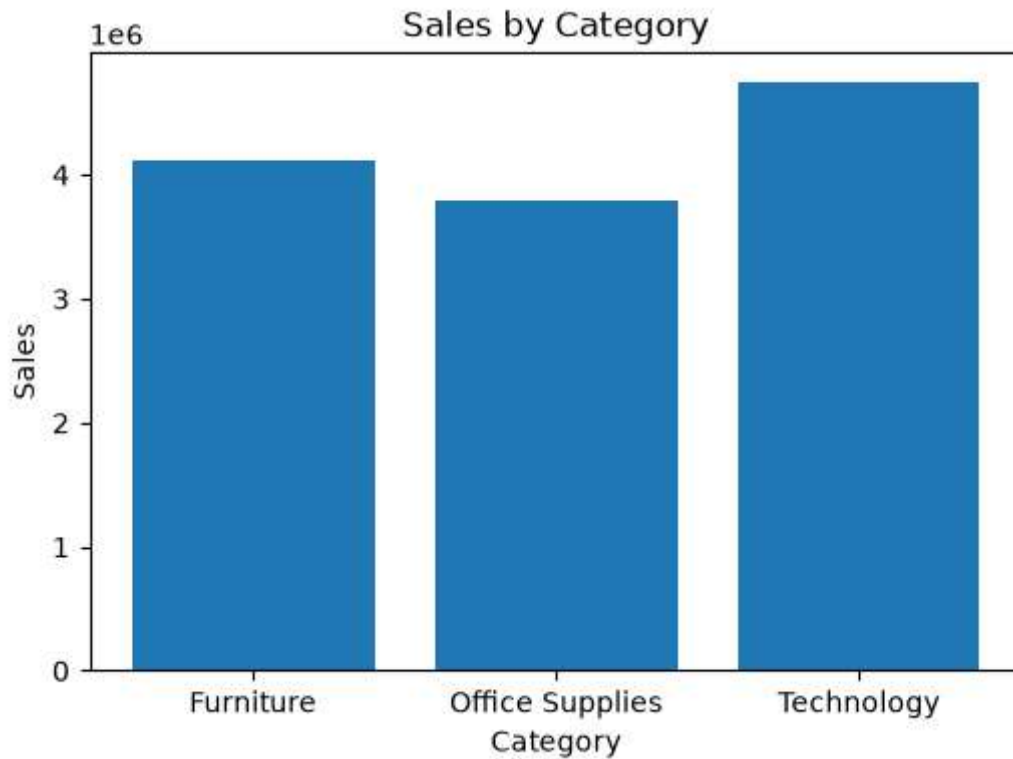
# Key Insights
```

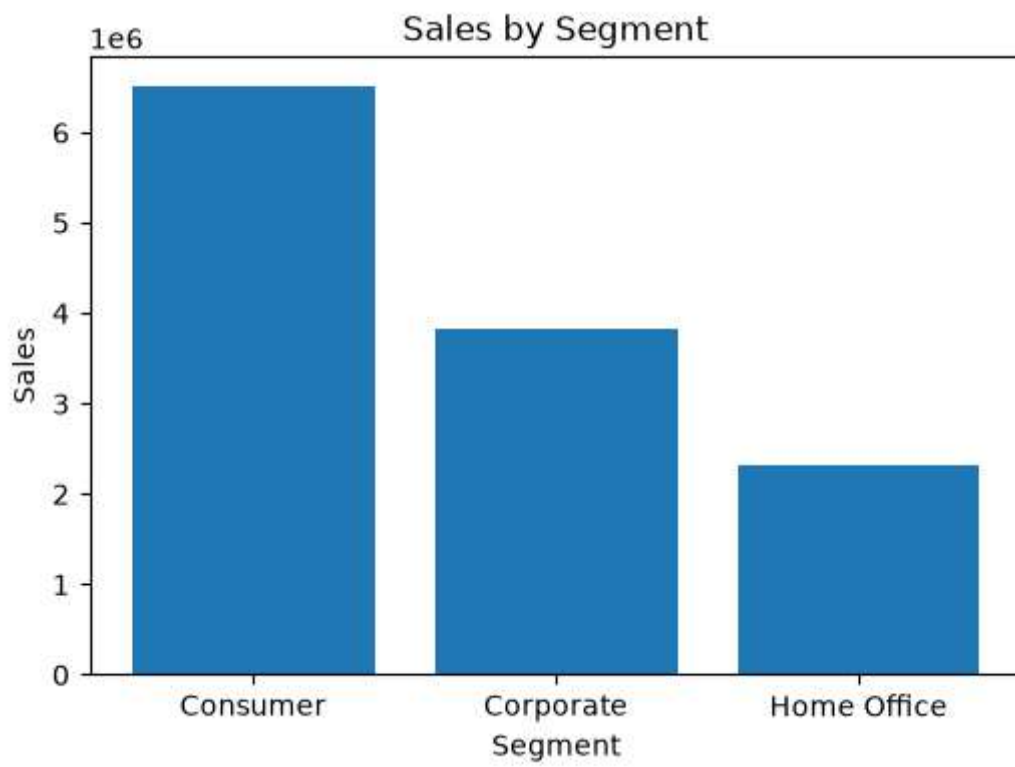
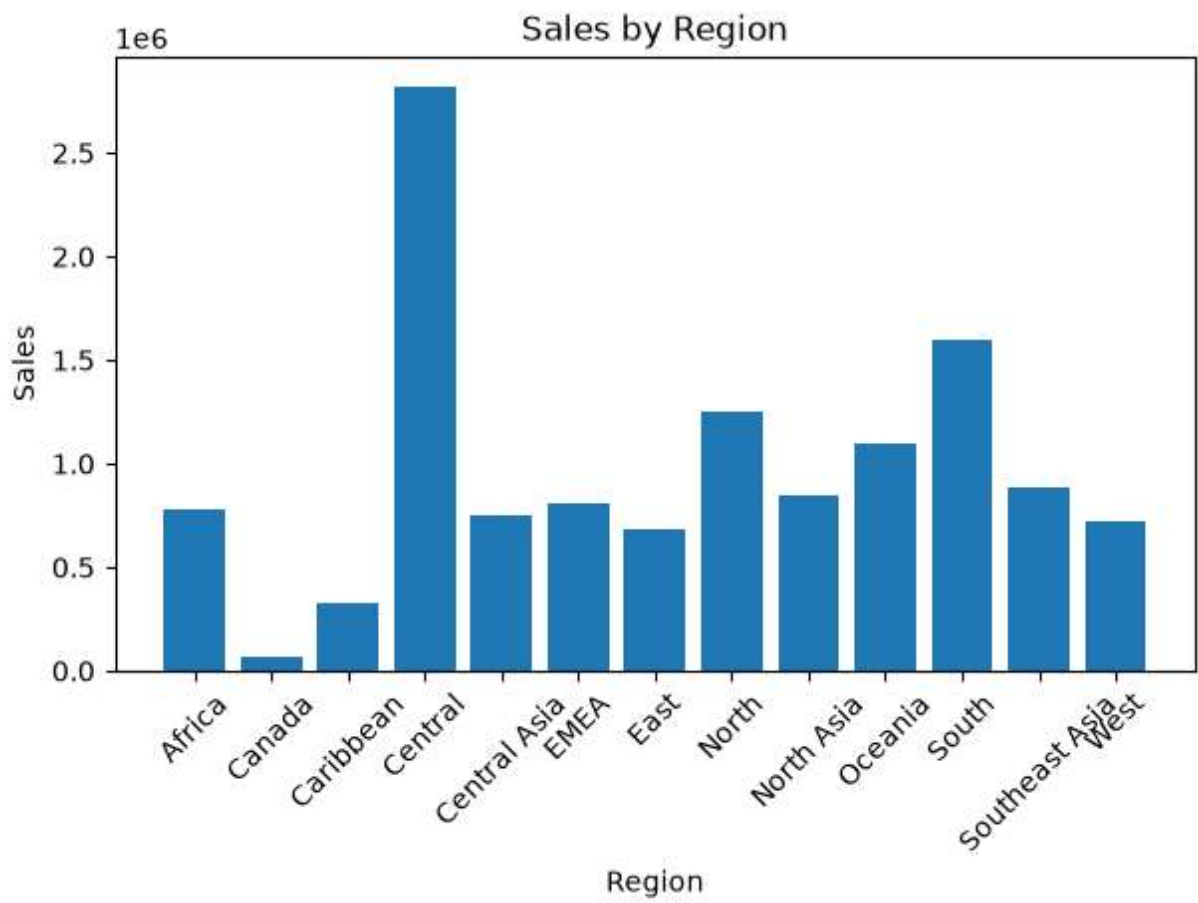
```
print("\n===== KEY INSIGHTS =====")
print("Total Sales:", round(df["Sales"].sum(), 2))
print("Total Profit:", round(df["Profit"].sum(), 2))
print("Top Category:", category_sales.idxmax())
print("Top Region:", region_sales.idxmax())
print("Top Segment:", segment_sales.idxmax())

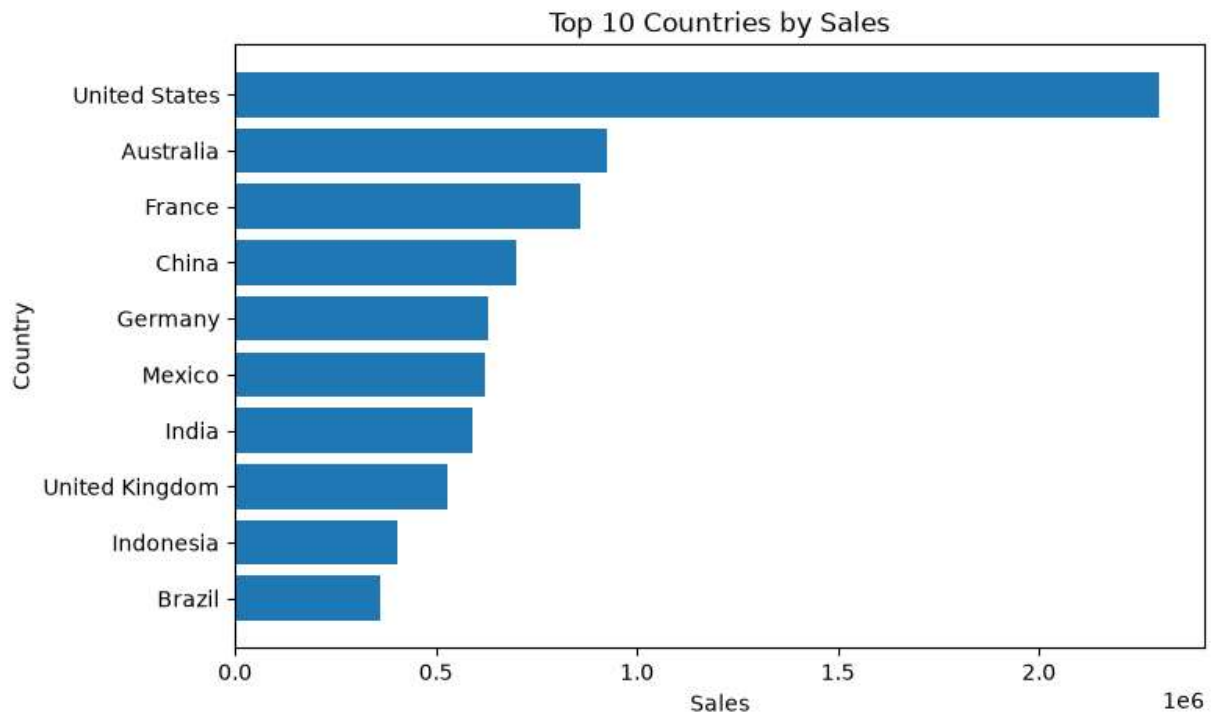
print("\nExperiment No. 9 Completed Successfully")
```

Dataset Loaded Successfully

Dataset Shape: (51290, 27)







===== KEY INSIGHTS =====

Total Sales: 12642905

Total Profit: 1467457.29

Top Category: Technology

Top Region: Central

Top Segment: Consumer

Experiment No. 9 Completed Successfully

```
In [ ]: # Experiment No. 9

## Experiment Name
Data Storytelling and Business Insight Generation using Interactive Visualizations

## Aim
To create a compelling data-driven story by analyzing a real-world dataset, develop

## Dataset Used
Netflix Movies and TV Shows Dataset

## Objective
To analyze Netflix content using data visualization and identify useful trends and
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QUESTIONS AND ANSWERS

Q1. What is data storytelling? How is it different from data visualization?

Answer: Data storytelling combines data, visualizations, and explanations to communicate insights. Data visualization mainly represents data through charts and graphs.

Q2. Why is storytelling important in business analytics and decision-making?

Answer: Storytelling makes complex data easy to understand and helps stakeholders identify trends and make informed decisions.

Q3. What are the essential components of an effective data story?

Answer: The main components are data, visualizations, insights, context, and a clear narrative.

Q4. How do Key Performance Indicators (KPIs) enhance business reporting?

Answer: KPIs provide important measurable values that help track performance and understand business results.

Q5. Why should visualizations be arranged in a logical sequence?

Answer: A logical sequence makes the analysis easier to follow and clearly explains the problem, findings, and conclusions.

Q6. What factors should be considered while selecting visualizations?

Answer: The type of data, purpose of analysis, audience, readability, and comparison requirements should be considered.

Q7. Explain how dashboards and storytelling complement each other in Business Intelligence.

Answer: Dashboards provide interactive visual information, while storytelling explains the meaning of the information and highlights important insights.

Q8. What challenges may arise while communicating analytical insights to non-technical stakeholders?

Answer: Common challenges include technical terms, complex charts, too much information, and difficulty understanding statistical results.

Q9. Give two real-world examples where data storytelling has influenced business or policy decisions.

Answer: Netflix uses viewing data to understand audience preferences and plan content. Governments use data stories to understand public trends and support policy decisions.

Q10. How can effective data storytelling improve strategic planning and organizational performance?

Answer: It helps organizations understand trends, identify opportunities and risks, and make better data-driven decisions.